Supplementary File - Safe and Stable Neural Network Dynamical Systems for Robot Motion Planning

1 Results and Benchmarks from Fig. 2 and Table I

1.1 Dataset Description

Dataset	No. of Demonstrations		**Data Size per Demonstration**
	Train	**Test**	
Sine	5	2	1000
S-Shape	5	2	1000
C-Shape	5	2	1000
Worm	5	2	1000
Angle	5	2	1000
G-Shape	5	2	1000
P-Shape	5	2	1000
N-Shape	5	2	1000
Sine (Fig. 6)	5	2	1000
3D C-Shape	10	5	432
2D Robot Dataset	3	2	12,432

Table 1: Training-Test Dataset Parameters for S^2-NNDS and ABC-DS methods

For *3D C-Shape* and *2D Robot Dataset*, we report the average number of points per demonstration, as this number may vary for each demonstration with robot data.

1.2 Initial Set Configurations

Dataset	Type	Expression
Sine	Rectangle	$-1.0 \leq x \leq -0.881, -0.039 \leq y \leq 0.072$
S-Shape	Rectangle	$0.681 \leq x \leq 0.823, 0.814 \leq y \leq 1.0$
C-Shape	Rectangle	$-0.048 \leq x \leq 0.070, 0.637 \leq y \leq 0.935$
Worm	Rectangle	$-1.0 \leq x \leq -0.940, -0.066 \leq y \leq -0.007$
Angle	Rectangle	$-1.0 \leq x \leq -0.837, -0.113 \leq y \leq 0.092$
G-Shape	Rectangle	$0.278 \leq x \leq 0.437, 0.496 \leq y \leq 0.784$
P-Shape	Rectangle	$-0.843 \leq x \leq -0.576, -0.683 \leq y \leq -0.560$
N-Shape	Rectangle	$-1.0 \leq x \leq -0.809, -0.902 \leq y \leq -0.843$
Sine (Fig. 6)	Rectangle	$-1.0 \leq x \leq -0.881, -0.0388 \leq y \leq 0.0716$
3D C-Shape	Cuboid	$-0.374 \leq x \leq 0.482$ $-0.782 \leq y \leq 0.474$ $-1.0 \leq z \leq -0.844$
2D Robot Dataset	Rectangle	$0.916 \leq x \leq 1.0, -0.059 \leq y \leq 0.059$

Table 2: Initial sets for S^2-NNDS approach.

Dataset	Type	Expression
Sine	Circle	$(x + 0.940)^2 + (y - 0.029)^2 \leq 0.0144$
S-Shape	Circle	$(x - 0.737)^2 + (y - 0.923)^2 \leq 0.0225$
C-Shape	Circle	$(x - 0.024)^2 + (y - 0.831)^2 \leq 0.04$
Worm	Circle	$(x + 0.966)^2 + (y + 0.038)^2 \leq 0.04$
Angle	Circle	$(x + 0.935)^2 + (y + 0.022)^2 \leq 0.04$
G-Shape	Circle	$(x - 0.36)^2 + (y - 0.644)^2 \leq 0.04$
P-Shape	Circle	$(x + 0.706)^2 + (y + 0.621)^2 \leq 0.0625$
N-Shape	Circle	$(x + 0.904)^2 + (y + 0.88)^2 \leq 0.04$
Sine (Fig. 6)	Circle	$(x + 0.940)^2 + (y - 0.029)^2 \leq 0.0144$

Table 3: Overapproximation of unsafe sets for ABC-DS approach.

1.3 Obstacle Configurations

Dataset	Type	Expression
Sine	Circle	$(x+0.4)^2 + (y-0.35)^2 \leq 0.01$
S-Shape	Rectangle	$0.3 \leq x \leq 1, 0.55 \leq y \leq 0.65$
C-Shape	Custom	$(\frac{x+0.3}{6})^{\frac{2}{3}} + (\frac{y-0.4}{3})^{\frac{2}{3}} \leq 0.15$
Worm	Rectangle	$-0.4 \leq x \leq -0.3, -0.1 \leq y \leq 0.05$
Angle	Rectangle	$-0.6 \leq x \leq -0.4, 0.7 \leq y \leq 1.0$
G-Shape	Circle	$x^2 + (y-0.25)^2 \leq 0.01$
P-Shape	Circle	$x^2 + (y-0.4)^2 \leq 0.04$
N-Shape	Custom	$\max(y, -4x - 0.8y - 1.6, -0.8y + 4x + 0.8) \leq 0$
Sine (Fig. 6)	Custom	$\max(y + 0.2\sin(9.5x - \pi/6) - 0.5,$ $-0.2\sin(9.5x - \pi/6) + 0.28 - y, x, -1 - x) \leq 0$
3D C-Shape	Sphere	$(x-0.4)^2 + (y-0.2)^2 + (z+0.3)^2 \leq 0.0625$
2D Robot Dataset	Multiple Circles	$(x-0.25)^2 + (y+0.2)^2 \leq 0.0036,$ $(x-0.8)^2 + (y+0.2)^2 \leq 0.0036,$ $(x-0.5)^2 + (y+0.55)^2 \leq 0.0036$

Table 4: Unsafe sets for S^2-NNDS method

Dataset	Type	Expression
Sine	Circle	$(x+0.4)^2 + (y-0.35)^2 \leq 0.01$
S-Shape	Ellipse	$\left(\frac{x-0.65}{0.4949}\right)^2 + \left(\frac{y-0.6}{0.07075}\right)^2 \leq 1$
C-Shape	Ellipse	$\left(\frac{x+0.3}{0.349}\right)^2 + \left(\frac{y-0.4}{0.174}\right)^2 \leq 1$
Worm	Ellipse	$\left(\frac{x+0.35}{0.0707}\right)^2 + \left(\frac{y+0.025}{0.1061}\right)^2 \leq 1$
Angle	Ellipse	$\left(\frac{x+0.5}{0.14145}\right)^2 + \left(\frac{y-0.85}{0.2121}\right)^2 \leq 1.0$
G-Shape	Circle	$x^2 + (y-0.25)^2 \leq 0.01$
P-Shape	Circle	$x^2 + (y-0.4)^2 \leq 0.04$
N-Shape	Ellipse	$\left(\frac{x+0.3}{0.1}\right)^2 + \left(\frac{y}{0.5}\right)^2 \leq 1$
Sine (Fig. 6)	Multiple Ellipses	$0.905x^2 - 0.659xy + 0.241y^2 + 1.890x - 0.793y + 1 \leq 0$ $84.668x^2 + 111.875xy + 42.789y^2 + 59.037x + 34.455y + 10.178 \leq 0$ $84.723x^2 - 111.720xy + 42.899y^2 + 90.270x - 64.252y + 23.968 \leq 0$ $104.406x^2 + 74.861xy + 49.915y^2 - 28.919x - 49.783y + 11.644 \leq 0$

Table 5: Overapproximation of unsafe sets for ABC-DS method.

1.4 Network and Function Structure

Dataset	**Dynamical System f_θ**		**Lyapunov Function $V_{\theta'}$**		**Barrier Function $B_{\theta''}$**	
	$\sigma(\mathbf{x})$	**Architecture**	$\sigma(\mathbf{x})$	**Architecture**	$\sigma(\mathbf{x})$	**Architecture**
Sine	Tanh	$64-64-64-64$	ELU	$128-128-128$	ELU	$128-128-128$
S-Shape	Tanh	$64-64-64-64$	ELU	$64-64-64-64$	ELU	$64-64-64-64$
C-Shape	Tanh	$32-32$	ELU	$64-64-64$	ELU	$64-64-64$
Worm	Tanh	$16-16$	ELU	$64-64-64$	ELU	$64-64-64$
Angle	Tanh	$16-16$	ELU	$64-64$	ELU	$64-64$
G-Shape	ELU	$64-64-64-64$	ELU	$128-128-128$	ELU	$128-128-128-128-128$
P-Shape	ELU	$64-64-64-64$	ELU	$128-128-128-128-128$	ELU	$128-128-128-128-128$
N-Shape	ELU	$64-64-64-64$	ELU	$128-128$	ELU	$128-128-128-128-128$
Sine (Fig. 6)	Tanh	$64-64-64-64$	ELU	$128-128-128$	ELU	$128-128-128-128$
3D C-Shape	Tanh	$128-128-128-128$	ELU	$128-128-128-128$	ELU	$128-128-128-128-128$
Robot dataset	Tanh	$64-64-64$	ELU	$128-128-128-128$	ELU	$128-128-128-128-128$

Table 6: NN architectures parameters for S^2-NNDS approach

Here, $\sigma(\mathbf{x})$ denotes the activation function in the NN.

Dataset	**Dynamical System**	**Lyapunov Function**	**Barrier Function**
Sine	5	2	3
S-Shape	5	4	4
C-Shape	5	4	2
G-Shape	5	4	4
P-Shape	5	4	2
N-Shape	7	6	4

Table 7: Polynomial Degrees for the functions optimized in ABC-DS

1.5 Training Time Comparison S^2-NNDS vs ABC-DS

Dataset	**S^2-NNDS**	**ABC-DS**
Sine	135.63s	348.70s
S-Shape	233.17s	1336.91s
C-Shape	128.20s	1226.97s
Worm	538.76s	FAIL
Angle	1011.63s	FAIL
G-Shape	247.17s	1248.40s
P-Shape	212.26s	1102.11s
N-Shape	626.65s	723.10s
Sine (Fig. 6)	534.41s	FAIL
3D C-Shape	752.56s	-
2D Robot Dataset	1554.34s	-

Table 8: Training Time for S^2-NNDS and ABC-DS approaches

1.6 Parameters for Verification using Conformal Prediction

Dataset	ϵ	β	N_{ver}
Sine	1e-4	3.5e-10	250,000
S-Shape	1e-4	4.3e-8	250,000
C-Shape	1e-4	4.3e-8	250,000
Worm	1e-4	4.3e-8	200,000
Angle	1e-4	0.0	200,000
G-Shape	1e-4	4.3e-8	150,000
P-Shape	1e-4	4.3e-8	200,000
N-Shape	1e-4	4.3e-8	200,000
Sine (Fig. 6)	1e-4	3.6e-10	250,000
3D C-Shape	1e-4	3.6e-10	250,000
2D Robot Dataset	1e-4	3.6e-10	250,000

Table 9: Conformal Prediction Parameters for S^2-NNDS approach

2 Results and Benchmarks from Table II

2.1 Dataset Description

Dataset	No. of Demonstrations		**Data size per Demonstration**
	Train	**Test**	
Worm	5	2	1000
P-Shape	5	2	1000
S-Shape	5	2	1000
Sine	5	2	1000

Table 10: Training-Test Dataset Parameters for S^2-NNDS and ABC-DS approaches

2.2 Initial Set Configurations

Dataset	**Type**	**Expression**
Worm	Circle	$(x+0.9650)^2+(y+0.0378)^2 \leq 0.01$
P-Shape	Circle	$(x+0.6789)^2+(y+0.5974)^2 \leq 0.0225$
S-Shape	Circle	$(x-0.5909)^2+(y-0.7403)^2 \leq 0.01$
Sine	Circle	$(x+0.9384)^2+(y-0.0288)^2 \leq 0.0049$

Table 11: Initial sets for S^2-NNDS & ABC-DS approaches

2.3 Obstacle Configurations

Dataset	**Type**	**Expression**
Worm	Custom	$202.721x^2+26.360y^2+31.097xy+200.694x+15.393y+48.672 \leq 0$
P-Shape	Custom	$67.499x^2+5.325y^2+20.201xy+43.52x+9.182y+6.482 \leq 0$
S-Shape	Custom	$12.261x^2+129.116y^2+20.605xy-21.331x-151.104y+45.089 \leq 0$
Sine	Custom	$597.983x^4-69.962x^3y+740.166x^2y^2-45.363xy^3+654.556y^4$ $+1077.284x^3-137.992x^2y+695.267xy^2+112.453y^3+598.803x^2$ $+89.649y^2-65.748xy+102.653x+0.579y+4.373 \leq 0$

Table 12: Unsafe sets for S^2-NNDS & ABC-DS approaches

2.4 Network and Function Structure

Dataset	DS Network f_θ		Lyapunov Function $V_{\theta'}$		Barrier Certificate $B_{\theta''}$	
	$\sigma(\mathbf{x})$	**Architecture**	$\sigma(\mathbf{x})$	**Architecture**	$\sigma(\mathbf{x})$	**Architecture**
Worm	Tanh	$16-16-16-16$	ELU	$128-128$	ELU	$128-128-128-128-128$
P-Shape	ELU	$64-64-64$	ELU	$128-128-128-128$	ELU	$128-128-128-128$
S-Shape	ELU	$128-128-128-128$	ELU	$64-64-64-64$	ELU	$64-64-64-64$
Sine	Tanh	$64-64-64$	ELU	$64-64-64$	ELU	$128-128-128-128$

Table 13: NN architectures parameters for S^2-NNDS approach

Here, $\sigma(\mathbf{x})$ denotes the activation function in the NN.

Dataset	**Dynamical System**	**Lyapunov Function**	**Barrier Function**
Worm	5	2	3
P-Shape	5	4	4
S-Shape	5	4	2
Sine	5	4	4

Table 14: Polynomial Degrees for the functions optimized in ABC-DS

2.5 Training Time Comparison S^2-NNDS vs ABC-DS

Dataset	S^2-NNDS	ABC-DS
Worm	246.77s	57.71s
P-Shape	1331.95s	100.40s
S-Shape	1172.84s	329.28s
Sine	2147.35s	1147.78s

Table 15: Training Time Comparisions for S^2-NNDS and ABC-DS approaches

2.6 Parameters for Verification using Conformal Prediction

Dataset	ϵ	β	N_{ver}
Worm	1e-4	4.32e-8	250,000
P-Shape	1e-4	6.14e-6	120,000
S-Shape	1e-4	4.32e-8	200,000
Sine	1e-4	4.32e-8	200,000

Table 16: Conformal Prediction Verification for S^2-NNDS approach